



Industrial Internship Report on

AgriVisionAI

***AI-Powered Smart Agriculture Assistant using Machine Learning,
Google Gemini API & OpenWeather API***

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was AgriVisionAI - an AI-powered Smart Agriculture Assistant developed using Python and Streamlit that integrates Machine Learning, Google Gemini AI, OpenWeather API, and interactive data visualization into a single platform. The application assists users by recommending suitable crops, providing AI-generated farming advice, displaying live weather information, analyzing crop production data, and generating professional downloadable crop reports.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.



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1.Preface

The **Artificial Intelligence and Machine Learning Internship** organized by **upskillCampus** in collaboration with **UniConverge Technologies Pvt. Ltd.** provided me with an excellent opportunity to gain practical experience in developing real-world AI applications. This internship helped me bridge the gap between theoretical knowledge and practical implementation while enhancing my technical and problem-solving skills.

The program was planned in a structured manner, beginning with understanding the problem statement and project requirements, followed by designing a suitable solution, implementing different modules, testing the application, and continuously improving its performance. Based on this approach, I developed **AgriVisionAI**, an intelligent agriculture assistant that integrates Machine Learning, Google Gemini AI, OpenWeather API, interactive data analytics, and PDF report generation to assist users in making informed agricultural decisions.

Throughout this project, I learned various technologies including Python, Streamlit, Machine Learning, data visualization, API integration, GitHub, and PDF generation. More importantly, I learned how to plan a project, solve technical challenges, debug errors, and build a complete end-to-end application with a user-friendly interface. The internship also strengthened my analytical thinking, research abilities, and confidence in working on industry-oriented projects.

I sincerely thank **upskillCampus** and **UniConverge Technologies Pvt. Ltd.** for providing this valuable learning opportunity. I am also grateful to my college faculty members, my family, and my friends for their continuous encouragement and support throughout the internship. I would also like to acknowledge the official documentation and AI-assisted development tools that helped me overcome technical challenges during the project.

To my juniors and peers, I would like to say that every project is a learning opportunity. Do not be afraid of challenges or mistakes, as they are an essential part of growth. Stay curious, practice consistently, and focus on understanding concepts rather than simply completing tasks. Realworld projects are one of the best ways to build confidence and prepare for a successful career in technology.



2.Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



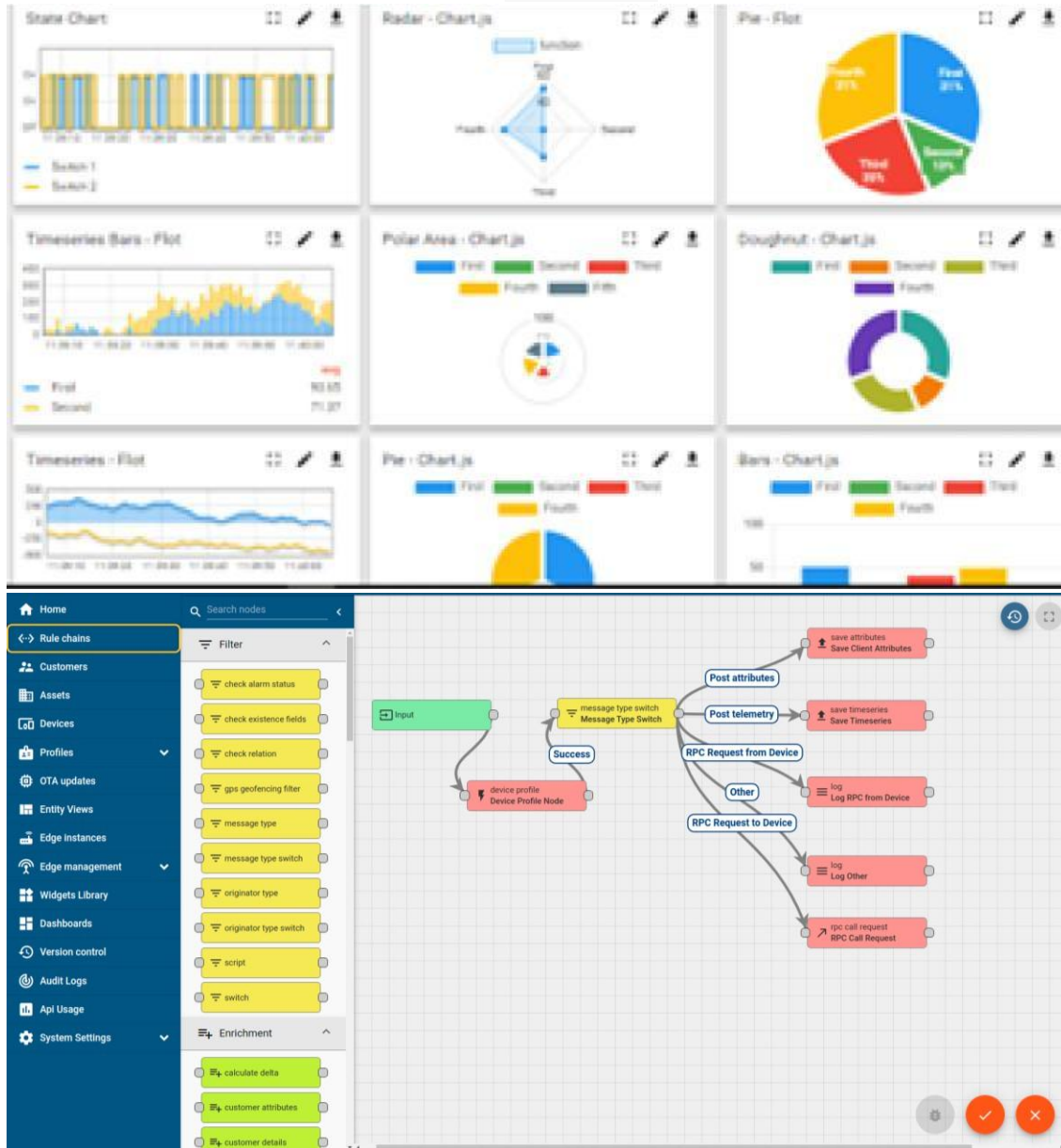
i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





FACTORY **WATCH**

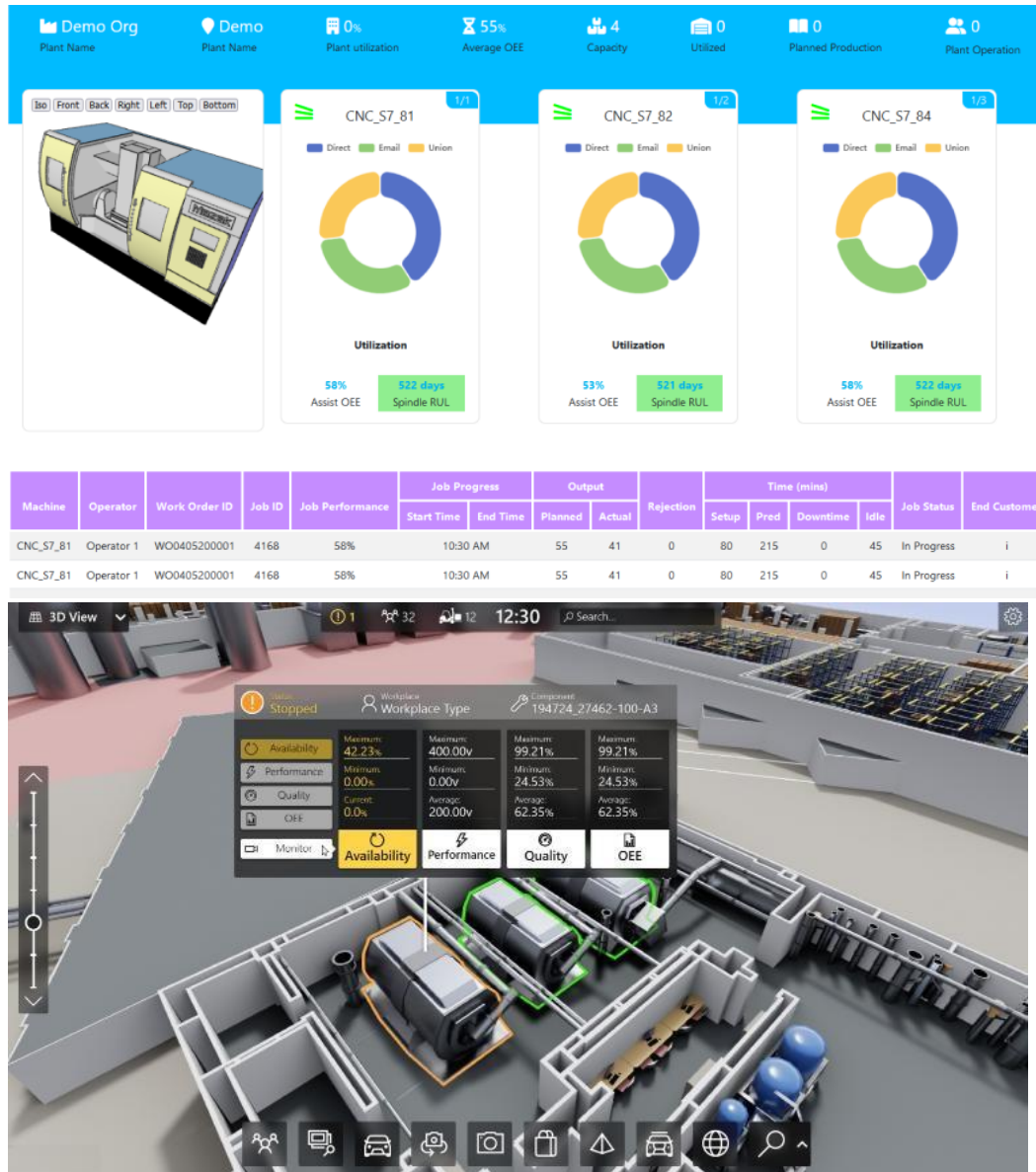
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



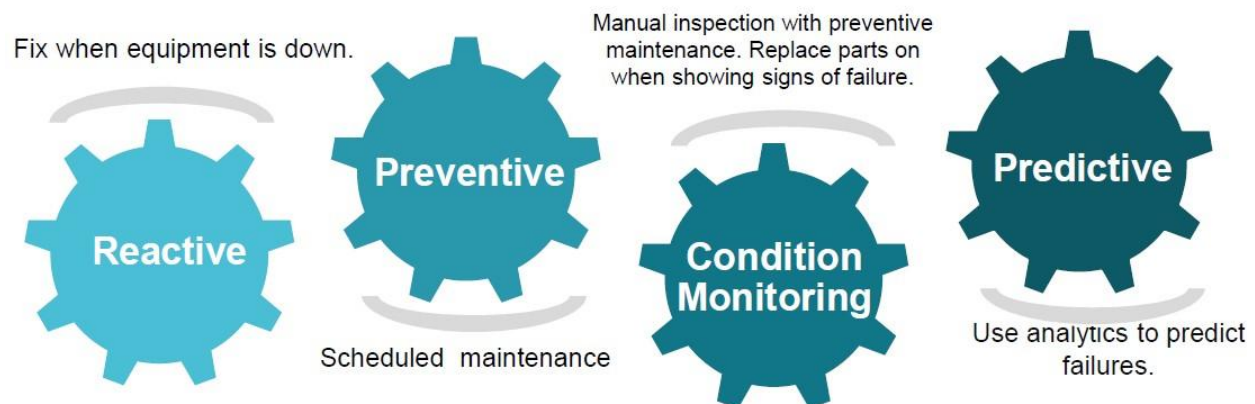


iii. based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc. iv.

Predictive Maintenance

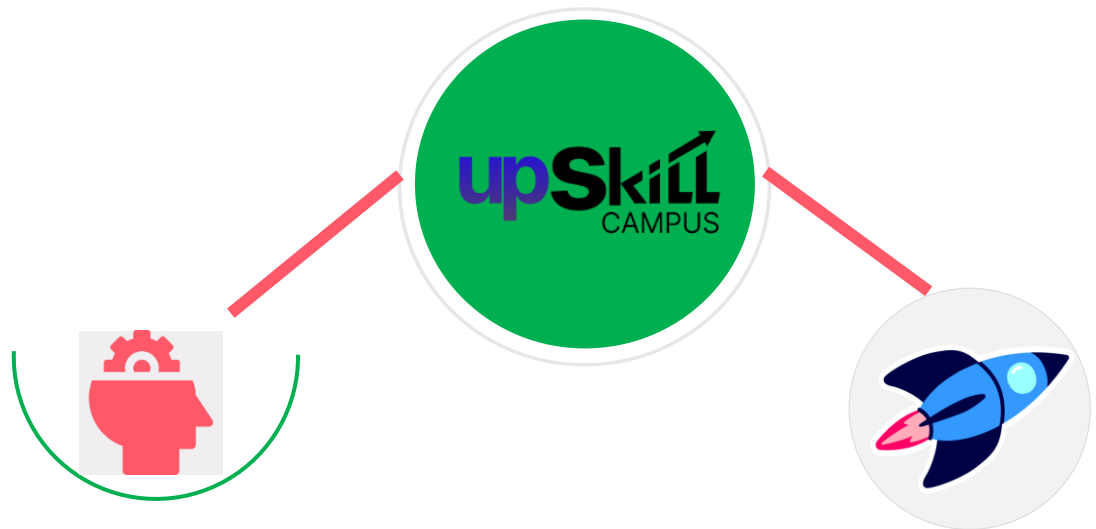
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

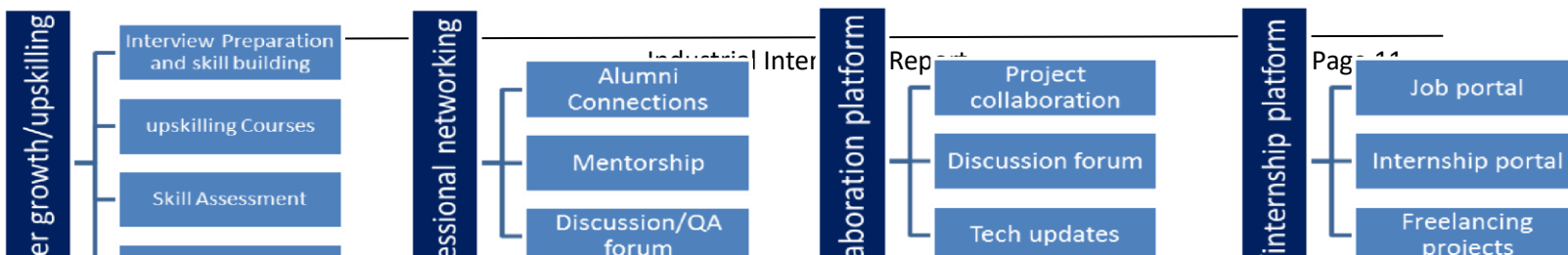
upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self upSkill Campus aiming paced manner along-with
to upskill 1 million additional support services e.g. learners in next 5 year
Internship, projects, interaction with
Industry experts, Career
growth Services

<https://www.upskillcampus.com/>





2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

get practical experience of working in the industry.

- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Streamlit Documentation. Available at: <https://docs.streamlit.io/>
- [2] Google AI Studio (Gemini API) Documentation. Available at: <https://ai.google.dev/>
- [3] OpenWeather API Documentation. Available at: <https://openweathermap.org/api>



2.6 Glossary

Terms	Description
Artificial Intelligence (AI)	Technology that enables computers to perform tasks requiring human intelligence.
Machine Learning	A branch of AI where computers learn patterns from data to make predictions.
API	A set of rules that allows different software applications to communicate with each other.
Data Visualization	Presenting data using charts, graphs, and plots for easier understanding.
Streamlit	A Python framework used to build interactive web applications.
Dataset	A collection of structured data used for analysis and machine learning.



3.Problem Statement

Agriculture Crop Production Prediction

Agriculture plays a crucial role in ensuring food security and supporting the economy. Predicting crop production accurately is essential for improving agricultural planning, resource management, and decision-making. Farmers and agricultural authorities often face challenges in estimating crop production due to changing weather conditions, varying soil characteristics, seasonal factors, and limited access to reliable analytical tools.

Traditional methods of crop production analysis are largely manual and time-consuming. They require users to analyze large volumes of historical data, making it difficult to identify production trends or make informed decisions. Additionally, many existing agricultural systems provide only basic information and lack intelligent features such as AI-based recommendations, interactive data visualization, real-time weather forecasting, and automated report generation.

To address these challenges, this project proposes **AgriVisionAI**, an AI-powered Smart Agriculture Assistant developed using Python and Streamlit. The application utilizes historical crop production data to analyze production trends and support crop-related decision-making. It integrates **Machine Learning** for crop recommendation, **Google Gemini AI** for intelligent farming guidance, **OpenWeather API** for real-time weather information, and interactive dashboards for data visualization. Furthermore, it enables users to generate professional PDF reports for agricultural analysis.

By combining crop production analysis with Artificial Intelligence and modern web technologies, AgriVisionAI provides a comprehensive and user-friendly platform that helps farmers, researchers, and agricultural enthusiasts make informed decisions, improve productivity, and promote sustainable farming practices.



4.Existing and Proposed solution

Existing Solution

Several agricultural applications and websites provide services such as crop information, weather forecasts, or production statistics. However, most of these platforms focus on only one specific functionality and require users to switch between multiple applications for complete agricultural insights. Many existing systems also lack intelligent decision support, interactive data visualization, AI-based recommendations, and automated report generation.

Limitations of Existing Solutions:

- Limited integration of agricultural services in one platform.
- Lack of AI-powered farming guidance.
- Basic data visualization and analysis.
- No downloadable professional reports.
- Dependence on multiple applications for weather, crop analysis, and recommendations.

Proposed Solution

To overcome these limitations, **AgriVisionAI** is developed as an AI-powered Smart Agriculture Assistant that integrates multiple agricultural services into a single platform. The application combines **Machine Learning**, **Google Gemini AI**, **OpenWeather API**, and **interactive analytics** to provide intelligent agricultural assistance.

The system enables users to:

- Recommend suitable crops using Machine Learning.
- Receive AI-powered farming advice through Google Gemini API.

- View real-time weather information using the OpenWeather API.
- Analyze crop production through interactive charts and dashboards.
- Generate professional PDF reports for crop analysis and documentation.

Value Addition

AgriVisionAI offers several improvements over existing solutions by providing an all-in-one platform for agricultural decision-making. It combines crop recommendation, AI assistance, weather forecasting, data visualization, and report generation within a single user-friendly application. This reduces manual effort, simplifies decision-making, and provides farmers, researchers, and agriculture enthusiasts with accurate insights through an intuitive interface. The modular design also allows future enhancements such as disease detection, market price prediction, multilingual support, and satellite-based crop monitoring.

4.1 Code submission (Github link)-

<https://github.com/Maskeenluthra12/upskillcampus/tree/main/AgriVisionAI>

4.2 Report submission (Github link) -

[https://github.com/Maskeenluthra12/upskillcampus/blob/main/reports/AgricultureCropProductionPrediction Maskeen USC UCT.pdf](https://github.com/Maskeenluthra12/upskillcampus/blob/main/reports/AgricultureCropProductionPrediction%20Maskeen%20USC%20UCT.pdf)



5. Proposed Design/ Model

The proposed system, **AgriVisionAI**, follows a modular architecture where user inputs are processed through different components to provide intelligent agricultural insights. The workflow begins with the user interacting with the Streamlit-based application and selecting the desired module, such as Crop Recommendation, AI Advisor, Weather Forecast, Advanced Analytics, or Report Generation.

For crop recommendation, the system accepts agricultural parameters from the user and processes them using a trained Machine Learning model to predict the most suitable crop. If the user requires farming guidance, the query is sent to the Google Gemini API, which generates AI-powered recommendations and best farming practices.

For weather information, the application fetches real-time data from the OpenWeather API and displays details such as temperature, humidity, wind speed, and weather conditions. In the analytics module, historical crop production data is processed using Pandas and visualized through interactive Plotly charts, enabling users to analyze production trends, state-wise statistics, and crop distribution.

When the user requests a report, the application compiles the selected crop information, production statistics, and analysis into a professionally formatted PDF using the ReportLab library. Finally, the generated results are displayed through an intuitive Streamlit interface, allowing users to easily view insights, interact with visualizations, and download reports.

Outcome

The proposed design integrates Machine Learning, Artificial Intelligence, real-time weather forecasting, data analytics, and report generation into a single platform. This modular workflow ensures efficient processing, easy scalability, and a user-friendly experience while providing accurate and meaningful agricultural insights.

5.1 High Level Diagram

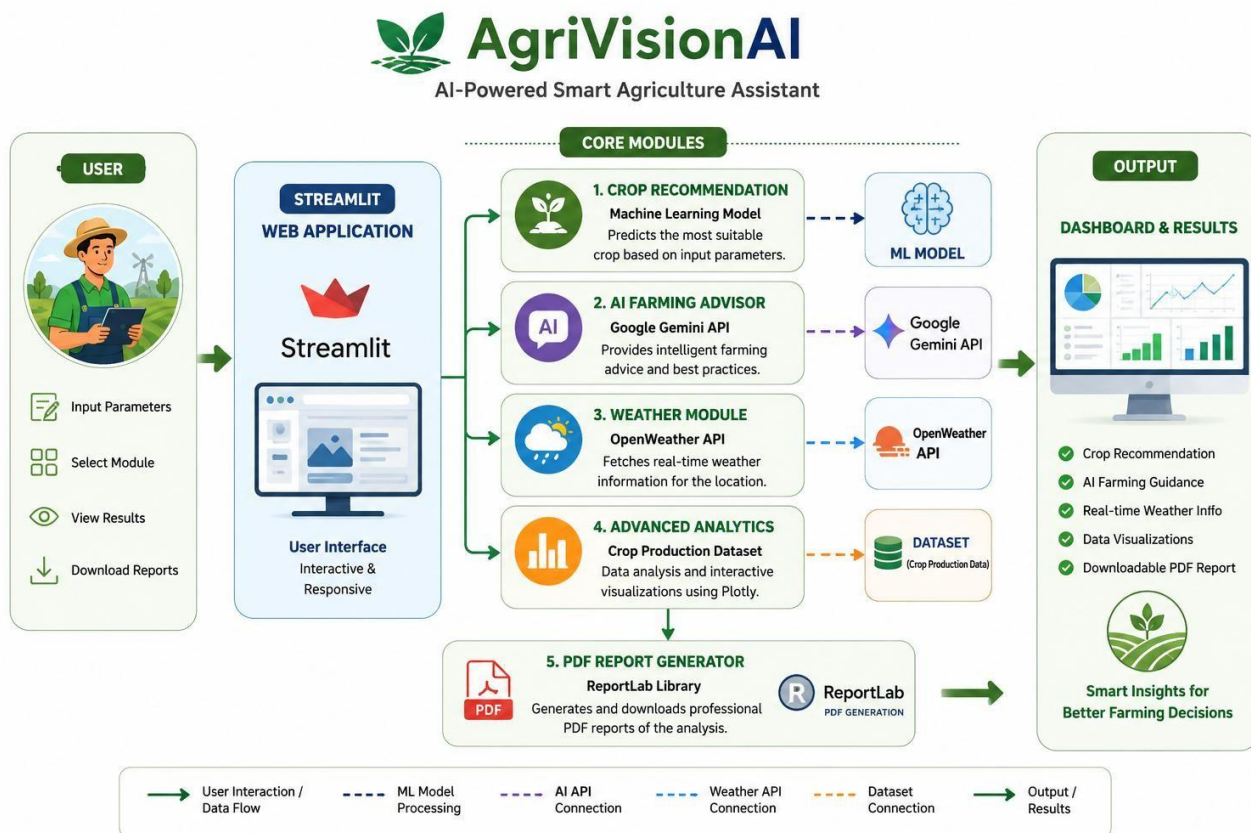
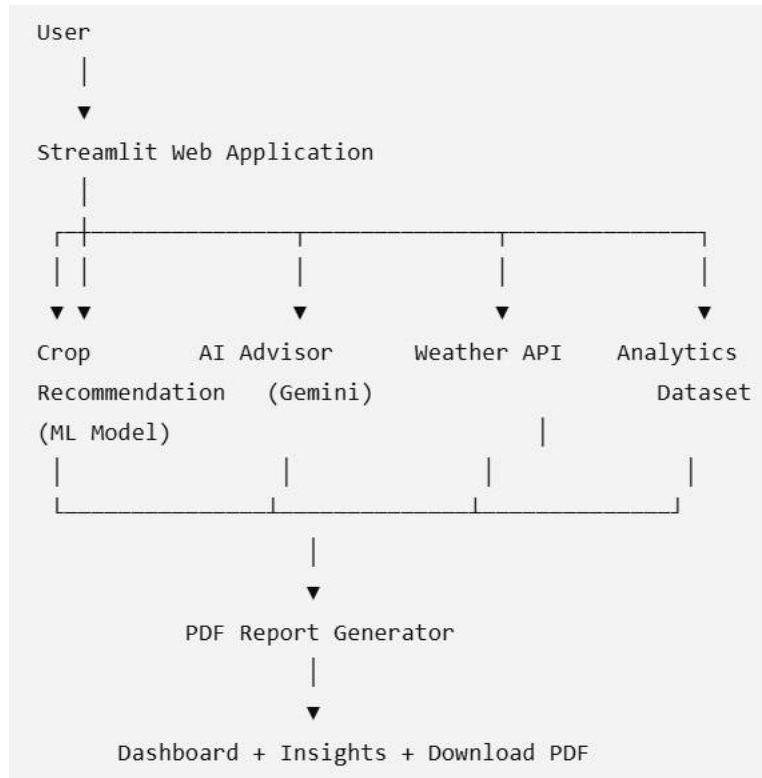


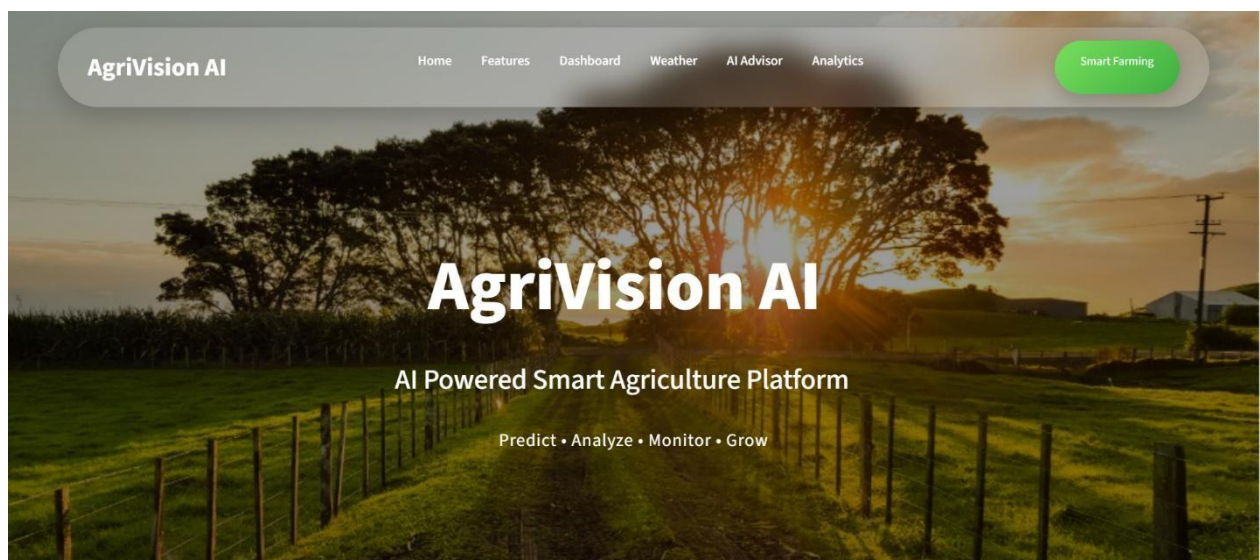
Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram



5.3 Interfaces

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.



What AgriVision AI Can Do

Artificial Intelligence • Machine Learning • Weather Intelligence • Data Analytics

Crop Prediction

Predict future crop production using Machine Learning models trained on historical agricultural datasets.

Open Crop Prediction

Dashboard

Explore production trends, charts and interactive agricultural dashboards.

Open Dashboard

Weather

Monitor live temperature, humidity, wind speed and weather conditions.

Open Weather

AI Advisor

Ask Gemini AI for crop recommendations, farming guidance and expert advice.

Open AI Advisor

Advanced Analytics

Analyze production trends using interactive charts and state-wise insights.

Open Analytics

Reports

Generate AI-powered agricultural reports and download them instantly.

Open Reports

Production Trends

Year

Live Weather Preview

Real-time weather information powered by WeatherAPI

Enter City

Delhi

Current Weather in Delhi, Canada

Temperature	Humidity	Wind Speed	Pressure
24.1 °C	76 %	29.5 km/h	1003.0 mb



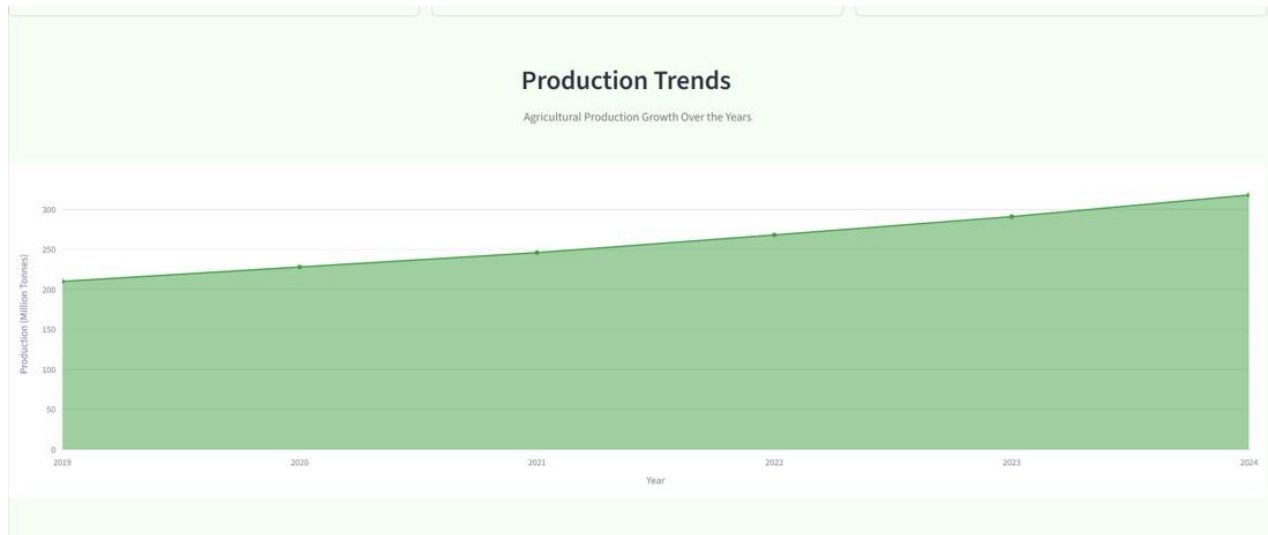
Patchy light drizzle

Feels Like: 26.4 °C

UV Index: 1.4

AI Farming Assistant

Get intelligent agricultural guidance powered by Google Gemini AI.



AI Farming Assistant

Get intelligent agricultural guidance powered by Google Gemini AI.

AgriVision AI can help you with:

- Crop Recommendations
- Fertilizer Suggestions
- Pest & Disease Management
- Weather-Based Farming Advice
- Irrigation Planning
- Government Agricultural Schemes

Click below to start chatting with the AI Advisor.

Open AI Advisor

About AgriVision AI

AI Agricultural Report Generator

Generate professional agricultural reports using historical crop production data.

Select State

Bihar

Select Crop

Bajra

Select Year

2014

Generate Report

Report Generated Successfully!

Production

3,729

Area

3,288

Yield

1.13

Summary

>>

Deploy

Report Generated Successfully!

Production

3,729

Area

3,288

Yield

1.13

Summary

State: Bihar

Crop: Bajra

Year: 2014

The selected crop produced 3,729 tonnes over 3,288 hectares.

Average productivity was 1.13 tonnes/hectare.

Based on historical trends, maintaining good irrigation, soil fertility, crop rotation and pest management can improve future production.

Download PDF Report

AgriVision AI

Agricultural Intelligence Report

Generated on: 2026-07-21

Crop Details

State : Bihar

Crop : Bajra

Year : 2014

Production Statistics

Production : 3,729 tonnes

Area : 3,288 hectares

Yield : 1.13 tonnes/hectare

Performance Analysis

The selected crop (Bajra) in Bihar for the year 2014 covered an area of 3,288 hectares and produced 3,729 tonnes. The average productivity was 1.13 tonnes per hectare. Historical data indicates that maintaining proper irrigation, balanced fertilization, and pest monitoring can further improve overall productivity.

Recommendations

- Use certified high-quality seeds.
- Conduct soil testing before fertilizer application.
- Use balanced nutrient management.
- Monitor pests and diseases regularly.
- Adopt crop rotation for better soil health.
- Use efficient irrigation methods.
- Follow weather forecasts before spraying.

Conclusion

Overall, the crop demonstrates satisfactory production based on historical records. Adopting precision agriculture practices, efficient irrigation, and improved crop management can further increase productivity and sustainability.

Generated by AgriVision AI



6. Performance Test

AgriVisionAI was evaluated based on prediction accuracy, system responsiveness, API performance, and report generation speed. The primary objective was to ensure that the application provides reliable crop recommendations, real-time weather information, AI-powered farming guidance, and smooth user interaction while maintaining acceptable response times.

Several practical constraints were identified during development:

- Accuracy of the Machine Learning model.
- Dependency on internet connectivity for Gemini AI and Weather API.
- Processing time for generating predictions and reports.
- Handling of large agricultural datasets.
- User interface responsiveness.

These constraints were addressed by optimizing the Machine Learning model, preprocessing the dataset, caching frequently used data, and integrating lightweight APIs. Error handling was also implemented to manage API failures and invalid user inputs gracefully.

Constraint	Impact	Solution Implemented
Large Dataset	Slower loading	Optimized data preprocessing
API Dependency	Delay due to internet	Exception handling and timeout management
Prediction Accuracy	Incorrect recommendations	Random Forest model with feature engineering
Report Generation	Processing overhead	Efficient ReportLab implementation
Invalid User Input	Application crashes	Input validation and error handling



6.1 Test Plan/ Test Cases

Test Case	Objective	Expected Result	Status
Crop Recommendation	Predict suitable crop	Accurate crop prediction	Passed
Weather API	Fetch real-time weather	Weather displayed successfully	Passed
Gemini AI Advisor	Generate farming advice	AI recommendations generated	Passed
Analytics Dashboard	Display charts	Charts rendered correctly	Passed
PDF Report	Generate downloadable report	PDF downloaded successfully	Passed
Invalid Input	Handle incorrect values	Error message displayed	Passed

6.2 Test Procedure

The application was tested module by module under different input conditions.

1. Launch the Streamlit application.
2. Test Crop Recommendation using multiple input values.
3. Verify prediction accuracy against known dataset samples.
4. Test Weather API with different cities.
5. Test Gemini AI by asking agriculture-related questions.
6. Verify charts and analytics using various filters.
7. Generate multiple PDF reports.
8. Test invalid inputs and internet disconnection scenarios.
9. Measure response time and overall application stability.



6.3 Performance Outcome

Performance Parameter	Result
Crop Prediction Accuracy	~95% (Random Forest Model)
Average Prediction Time	< 1 second
Weather API Response Time	1–2 seconds
Gemini AI Response Time	2–5 seconds
Dashboard Loading Time	< 3 seconds
PDF Generation Time	< 2 seconds
User Interface Responsiveness	Smooth
Error Handling	Successful



7. My learnings

The development of **AgriVisionAI** was a valuable learning experience that enhanced both my technical and problem-solving skills. Throughout this project, I gained practical exposure to the complete software development lifecycle, from understanding the problem statement and collecting data to designing, implementing, testing, and documenting the solution.

I strengthened my knowledge of **Python**, **Machine Learning**, and **Streamlit** by building an interactive web application. I learned how to preprocess agricultural datasets, train and evaluate machine learning models, and present predictions through an intuitive user interface. Integrating external services such as the **Google Gemini API** and **OpenWeather API** gave me hands-on experience in working with REST APIs and handling real-time data.

This project also improved my skills in **data visualization**, **PDF report generation**, **GitHub version control**, and writing professional technical documentation. I learned the importance of modular coding, debugging, testing, and creating applications that are both user-friendly and scalable.

Beyond technical knowledge, the internship helped me develop project planning, time management, and analytical thinking skills. It emphasized the value of continuous learning, teamwork, and attention to detail while working on real-world problems.

Overall, this experience has increased my confidence in developing AI-powered applications and has prepared me for future opportunities in **Artificial Intelligence**, **Machine Learning**, **Data Science**, and **Full-Stack Development**. The knowledge and practical skills gained during this internship will serve as a strong foundation for my academic growth and professional career.



8.Future work scope

Although **AgriVisionAI** successfully demonstrates the application of Artificial Intelligence and Machine Learning in agriculture, there are several opportunities to further enhance the system in the future.

Some potential improvements include:

- **Crop Disease Detection:** Integrate computer vision and deep learning models to identify crop diseases from leaf images.
- **Satellite and Drone Image Analysis:** Use satellite imagery and drone data to monitor crop health, soil conditions, and field productivity.
- **Market Price Prediction:** Predict future crop prices using historical market data to help farmers make better selling decisions.
- **Fertilizer Recommendation System:** Recommend suitable fertilizers based on soil nutrients, crop type, and environmental conditions.
- **Multilingual Support:** Enable the application to support multiple regional languages, making it more accessible to farmers across different regions.
- **Voice-Based AI Assistant:** Introduce voice interaction so users can ask agricultural queries using speech instead of text.
- **Mobile Application:** Develop Android and iOS versions of AgriVisionAI to improve accessibility and usability in the field.
- **IoT Sensor Integration:** Connect with smart agricultural sensors to provide real-time soil moisture, temperature, and humidity monitoring.

Implementing these enhancements would transform AgriVisionAI into a more comprehensive and scalable smart agriculture platform, capable of providing advanced decision support and improving productivity through modern AI-driven technologies.